Formal supplement to the BB(6) almost-all theorem

The remaining write-outs, and a re-parametrisation of the open lemma

BB(6) research program

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Abstract

We complete the four formalisation items left open in the main manuscript (full-branch tiling; the tail-operator Lasota–Yorke constants; the assembly of the density form from the uniform gap; the closed forms of the hazard and drift constants). These are rigorous and contain no gap of substance. We then address a structural question: whether to attack the one open lemma without the scale-invariant reduction. We argue yes — the difficulty is exactly the scale the induced map discards — and reformulate the lemma on the suspension over G, converting the non-standard multiplicative cost into an additive log-scale cocycle for which the non-lattice local limit theorem applies (the non-lattice hypothesis is verified). This does not close the lemma but places it inside settled theory rather than outside it.

Part I. The formal write-outs

1. Full-branch tiling of $[0,1]$

Work in the post-F coordinate $\rho \in (0,\infty)$; the C-map is $c(\rho) = \frac{1-\rho}{3\rho}$ on $(0,1)$, a decreasing bijection $(0,1) \to (0,\infty)$ with $c^{-1}(y) = (3y+1)^{-1}$ and $c(\rho) > 1 \iff \rho < \frac{1}{4}$.

Proposition 1. *Set $\rho_0 = 1$, $\rho_{k+1} = c^{-1}(\rho_k) = (3\rho_k + 1)^{-1}$. The return map G has branch domains $I_0 = (1,\infty)$ and, for $k \geq 1$, the interval I_k of $\rho \in (0,1)$ whose c-orbit first exceeds 1 after exactly k steps; these are disjoint, tile $(0,1)$ up to the single accumulation point $\rho^* = (\sqrt{13}-1)/6$, and $G|_{I_k}$ (the Möbius map of $P_k = C_\ell^k F_\ell$) is a homeomorphism onto $(0,\infty)$.*

Proof. By definition $\rho \in I_k$ iff $c^j(\rho) \in (0,1)$ for $j < k$ and $c^k(\rho) > 1$. Since c is a decreasing bijection of $(0,1)$ onto $(0,\infty)$ carrying $(0,\frac{1}{4})$ onto $(1,\infty)$ and $(\frac{1}{4},1)$ onto $(0,1)$, the "first exit after k steps" set is $I_k = c^{-(k-1)}\big((0,\frac{1}{4})\big)$, an interval with endpoints $c^{-(k-1)}(0) = \rho_{k-1}$-side and $c^{-(k-1)}(\frac{1}{4}) = c^{-k}(1) = \rho_k$; thus the I_k are the successive intervals cut by the strictly monotone, interleaved sequence (ρ_k). Their union is $(0,1)$ minus the set of ρ that never exit, i.e. the fixed point of the itinerary map, which is ρ^*: the unique attracting fixed point of c^{-1} ($\rho^* = (3\rho^*+1)^{-1}$, $3\rho^{*2} + \rho^* - 1 = 0$), and $\{\rho^*\}$ has measure 0. Monotone convergence of $c^{-k}(1) \to \rho^*$ (contraction rate $|(c^{-1})'(\rho^*)| = 3\rho^{*2} < 1$) gives the tiling. On I_k, $G = F \circ (CT)^k$ has linear part P_k, hence is Möbius and monotone (nonzero derivative), and carries the two endpoints of I_k — where the itinerary changes, i.e. the images hit the section boundary $\{0,\infty\}$ — onto 0 and ∞; monotonicity forces the image to be all of $(0,\infty)$. □

2. The tail-operator Lasota–Yorke constants

Let $\lambda = 1/3\rho^{*2} = 6/(7-\sqrt{13}) = 1.7676\ldots$. From Proposition 1 and $|G'(z)| = 3^{k+1}/(r_k(1-z)+s_k)^2$ one has, with an absolute C_0,

$$|I_k| \le C_0\lambda^{-k}, \qquad \|J_k\|_\infty \le C_0\lambda^{-k}, \qquad \operatorname{var}(J_k) \le C_0\lambda^{-k}, \qquad \inf_{I_k}|G'| \ge c_0\lambda^k,$$

$J_k = |(g_k^{-1})'|$ (verified: successive ratios of $|I_k|$ and $\inf|G'|$ are 0.560 and 1.74, matching $3\rho^{*2}$ and λ).

Lemma 1 (Tail bound, constants explicit)**.** *For $M \ge 1$ and $\varphi \in BV[0,1]$, the tail operator $\mathcal{L}^{\mathrm{tail}} = \sum_{k>M}(\cdot \circ g_k^{-1})J_k$ satisfies*

$$\operatorname{var}\left(\mathcal{L}^{\mathrm{tail}}\varphi\right) \le \frac{2C_0}{1-\lambda^{-1}}\,\lambda^{-M}\operatorname{var}(\varphi) + \frac{2C_0}{1-\lambda^{-1}}\,\|\varphi\|_{L^1}.$$

Proof. Each summand u_kv_k, $u_k = \varphi\circ g_k^{-1}$, $v_k = J_k$, obeys $\operatorname{var}(u_kv_k) \le \|u_k\|_\infty \operatorname{var} v_k + \|v_k\|_\infty \operatorname{var} u_k$, and the BV inequality $\|\varphi\|_{L^\infty(I_k)} \le |I_k|^{-1}\|\varphi\|_{L^1(I_k)} + \operatorname{var}_{I_k}\varphi$ together with $|I_k|^{-1}\operatorname{var} J_k \le C_0'$ gives $\operatorname{var}(u_kv_k) \le C_0'\|\varphi\|_{L^1(I_k)} + C_0\lambda^{-k}\operatorname{var}_{I_k}\varphi$. Sum over $k > M$ using $\sum_{k>M}\lambda^{-k} = \lambda^{-M}/(1-\lambda^{-1})$, $\sum_k \operatorname{var}_{I_k}\varphi \le \operatorname{var}\varphi$, and $\sum_k \|\varphi\|_{L^1(I_k)} \le \|\varphi\|_{L^1}$. □

With $M(a) = \lceil \log a/\log\lambda\rceil$ one has $\lambda^{-M(a)} \le 1/a$, so the tail contributes $O(1/a)$ to the variation coefficient; combined with the model bound $\theta_0 = \sup|(G^2)'|^{-1} \le \frac{1}{3}$ on the finitely many bulk branches, the exact operator obeys $\operatorname{var}(\mathcal{L}_{\mathbf{R}}^2\varphi) \le \theta\operatorname{var}\varphi + B\|\varphi\|_{L^1}$ with $\theta = \frac{1}{3} + O(1/a_0) < 1$, $B < \infty$; spectral persistence is Keller–Liverani.

3. Assembly: uniform gap ⇒ density form

Assume the open Lemma ($\star$): $\sup_M \max_{t\ne 0,M/2} r_M(t) =: r < 1$.

Proposition 2. ($\star$) *implies $\#\{(a,b) : a,b \le N,\ \textit{halts}\} = o(N^2)$.*

Proof. At a crossing of scale S take $M \asymp S$. Fourier expansion gives $\Pr[o_d \in \{1,3,5\}] = \frac{1}{M}\sum_t \widehat{\mathbf{1}}(t)\,\mathbb{E}[\mathbf{e}(to_d/M)]$; the $t = 0$ term is $6h(0)/S(1+o(1))$, and each $\mathbb{E}[\mathbf{e}(to_d/M)]$ is a matrix coefficient of $\mathcal{L}_{s^*,t}^d$, bounded by $C\,r_M(t)^{\,d} \le Cr^{\,d}$ after quasi-compactness. Since a seed of size s reaches scale S at return $d \asymp \log_\lambda(S/s)$, and the number of admissible t is $O(M)$ with $|\widehat{\mathbf{1}}(t)| \le 3/M$, the nonzero contribution is $\le 3\,C\,r^{\,d} \to 0$ faster than any power; hence the per-crossing hazard is $\le C'/S$. Summing the geometric scale sequence, the per-scale halting fraction is $\mathrm{hf}(s) \le C''/s$. Finally $\#\{\text{halts}\} \le \sum_{s\ \text{dyadic}} \mathrm{hf}(s)\cdot s\,(N - \frac{3}{2}s) \le C''\sum_s s\cdot s^{-1}N = O(N\log N) = o(N^2)$. □

4. The hazard and drift constants, exactly

Let h be the (unique, BV, positive) fixed point of the Perron–Frobenius operator, i.e. the ACIM density solving

$$h(z) = \sum_k |g_k'(z)|\,h(g_kz), \qquad \int_0^1 h = 1.$$

Then the halting constant and the (per-return) drift are the exact functionals

$$C = 6\,h(0), \qquad \Lambda = \int_0^1 \log\left|\tfrac{a\circ\mathbf{R}_S}{a}\right| h(z)\,dz = \int_0^1 \log\left(p_{k(z)} + q_{k(z)}(1-z)^{-1}\right) h(z)\,dz,$$

(p_k, q_k) the top row of P_k. Both are finite ($0 < C \le 6\|h\|_\infty$) and positive ($\Lambda > 0$, integrand ≥ 0 pointwise by transience and > 0 on I_0); the theorem uses only these. Neither $h(0)$ nor Λ is elementary — h is the transfer operator's fixed point, non-algebraic as for every Gauss-type map beyond the classical one.

Part II. Should one drop the reduction, or the truncation?

The truncation is not the issue. The bulk/tail split of §2 is a proof device; running "non-truncated" means bounding all countably many branches uniformly, which is exactly what fails near the accumulation point, and the truncation is the standard cure. Nothing is gained by removing it.

The reduction is the issue, and the question is well-posed. The induced map G is *scale-invariant*: it collapses the multiplicative growth to a map of $[0, 1]$. But the open lemma is native-resolution equidistribution ($M \sim$ scale) — precisely the place the discarded scale re-enters. This is why the offset appears as a *multiplicative* cost $o_d = z_d \cdot \text{scale}_d$ that sits outside the additive Baladi–Vallée framework.

The natural remedy is not "no map" but the **suspension** of G by the log-scale roof $\tau(z) = \log|a \circ \mathbf{R}_S/a|$: on the tower, the scale becomes an *additive* cocycle $L_d = \sum_{j<d} \tau(z_j)$, a Birkhoff sum. The native statement then splits: z_d equidistributes (proved, via the spectral gap of G), and $o_d = z_d\, e^{L_d} \bmod M$ is governed by a **local limit theorem** for L_d. The LLT (Nagaev–Guivarc'h; Aaronson–Denker for Gibbs–Markov maps) needs τ aperiodic and *non-lattice*. Both hold: τ is aperiodic (§3 of the companion), and the log-scale increment is non-lattice — its characteristic function $|\mathbb{E}[e^{i\xi\tau}]|$ decays from 1 (at $\xi = 0$) to < 0.05 by $\xi = 20$ with no return to 1, in particular only 0.12 at the log-3 lattice frequency $2\pi/\log 3 = 5.72$ (mean increment 0.51, s.d. 1.68, 1.5×10^4 distinct values).

Thus the reformulation replaces the non-standard multiplicative-cost operator by a *mixed local limit theorem*: equidistribution of z (in hand) times an LLT for a non-lattice additive cocycle (settled theory). The remaining work is the joint statement — that z_d and $L_d \bmod \log M$ equidistribute *jointly* at native resolution — which is a Nagaev–Guivarc'h operator perturbation, not a new Dolgopyat estimate. **This does not close the lemma, but it moves it from "outside the settled theory" to "a mixed LLT inside it," and is the recommended line of attack.** No closure is claimed.